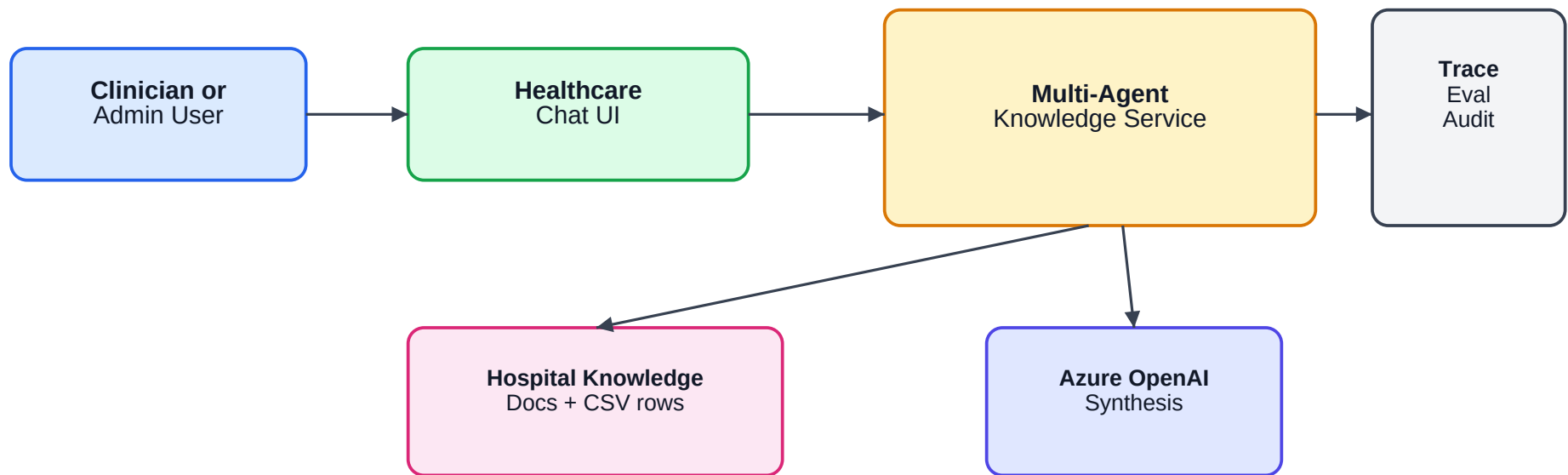


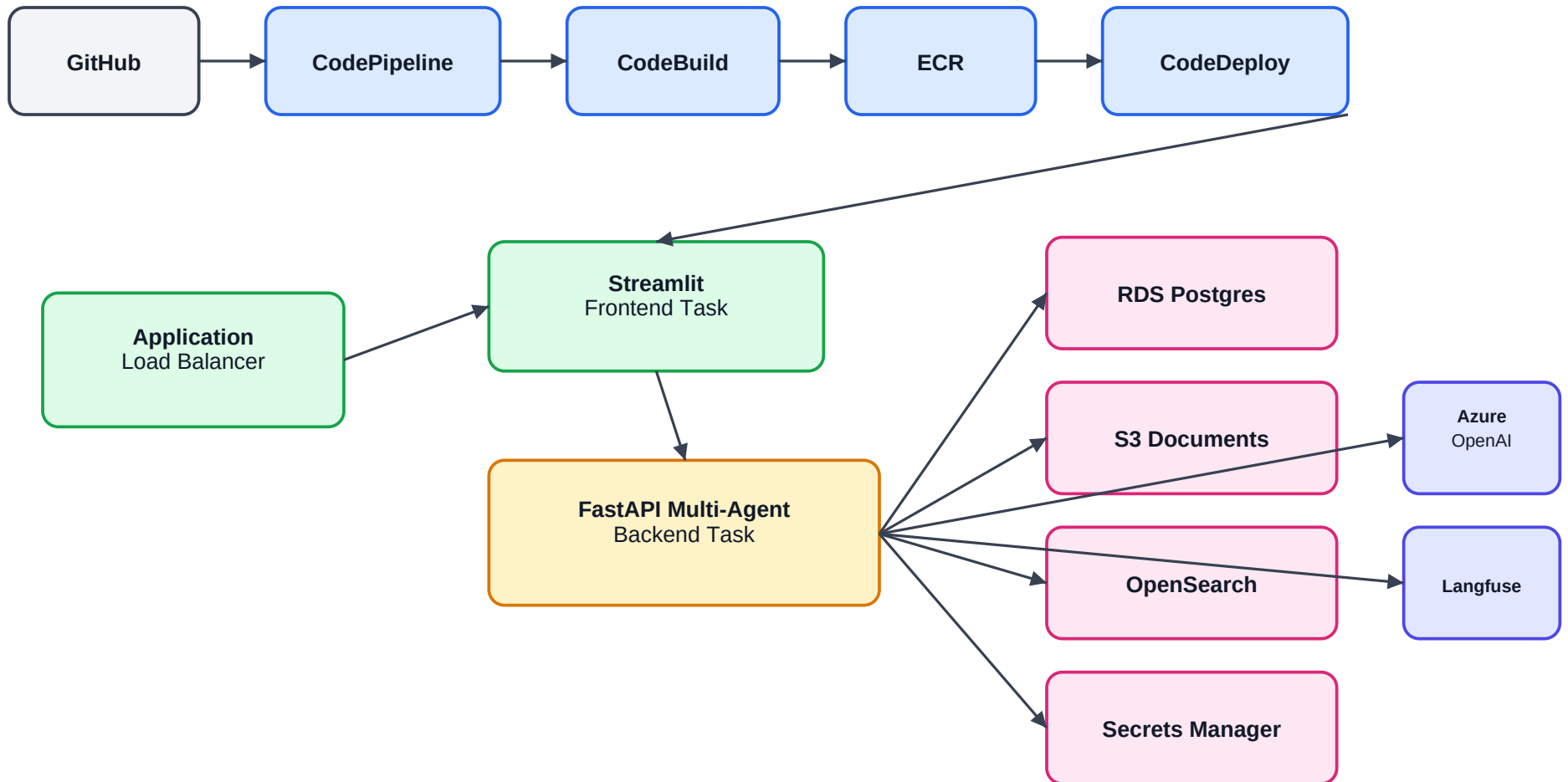
# HLD 1: Stakeholder View

Project: dstrmaysam-healthcare-knowledge-multi-agent



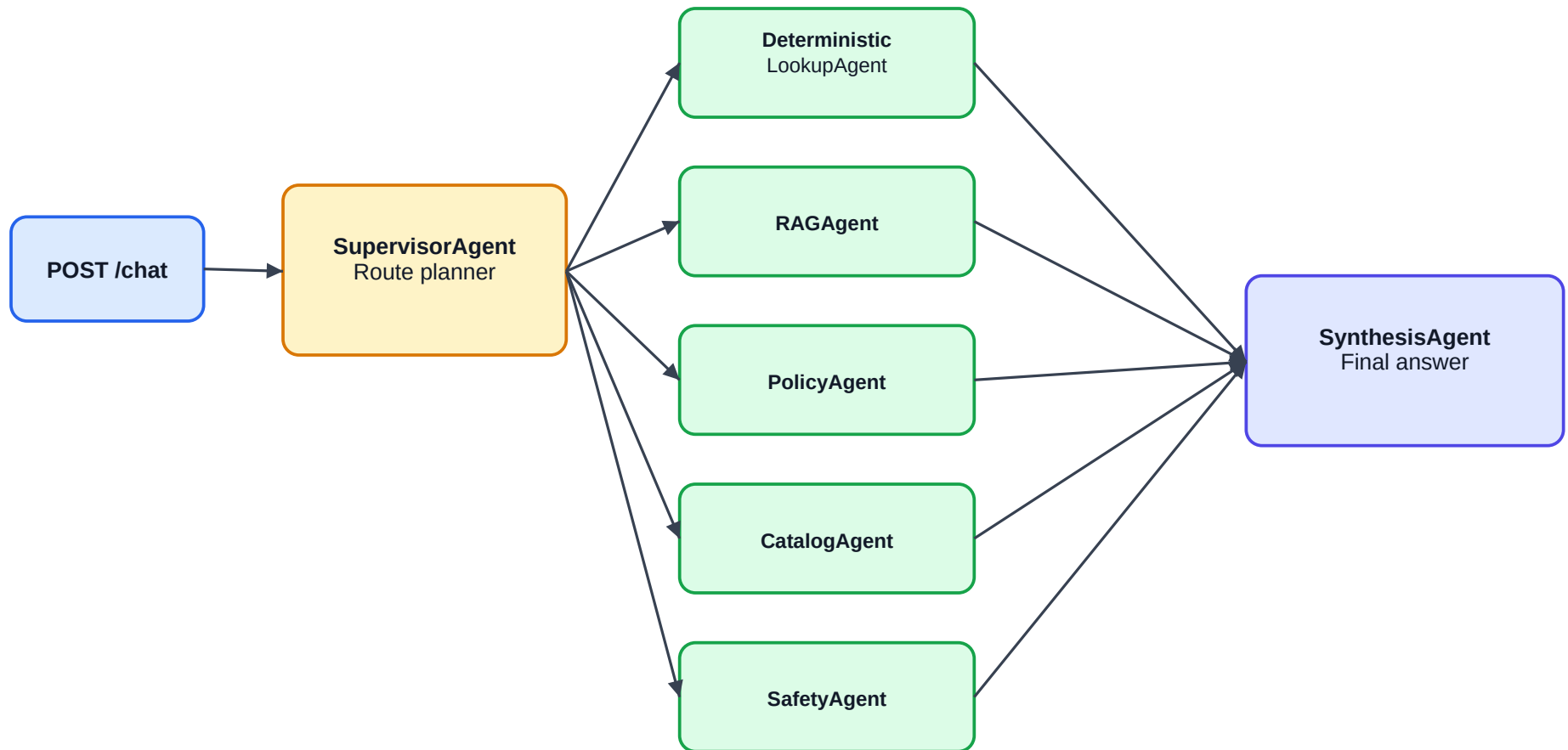
Users ask questions in one chat interface. The backend supervisor decides which specialist agents to call, gathers evidence from documents and structured rows, and returns a consistent answer with audit metadata.

## HLD 2: Technical AWS View



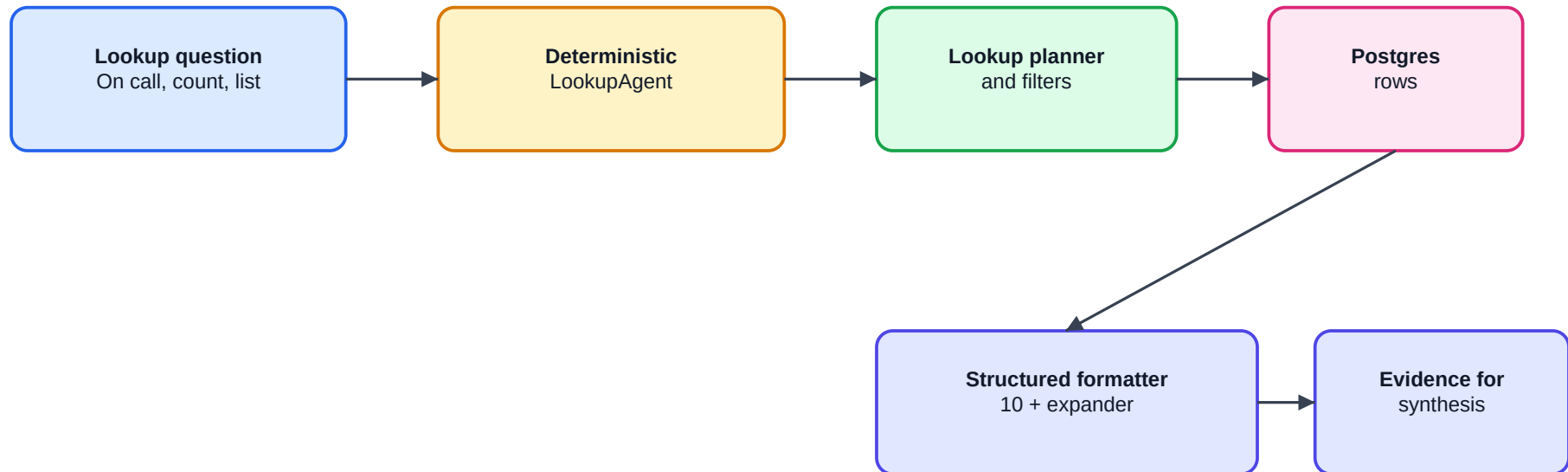
GitHub source flows through CodePipeline, CodeBuild, ECR, and CodeDeploy into ECS Fargate. Runtime dependencies are S3, OpenSearch Serverless, Secrets Manager, RDS PostgreSQL, Azure OpenAI, Langfuse, and CloudWatch Logs. DynamoDB is not used in the new deployment path.

# LLD 1: Online Chat Request



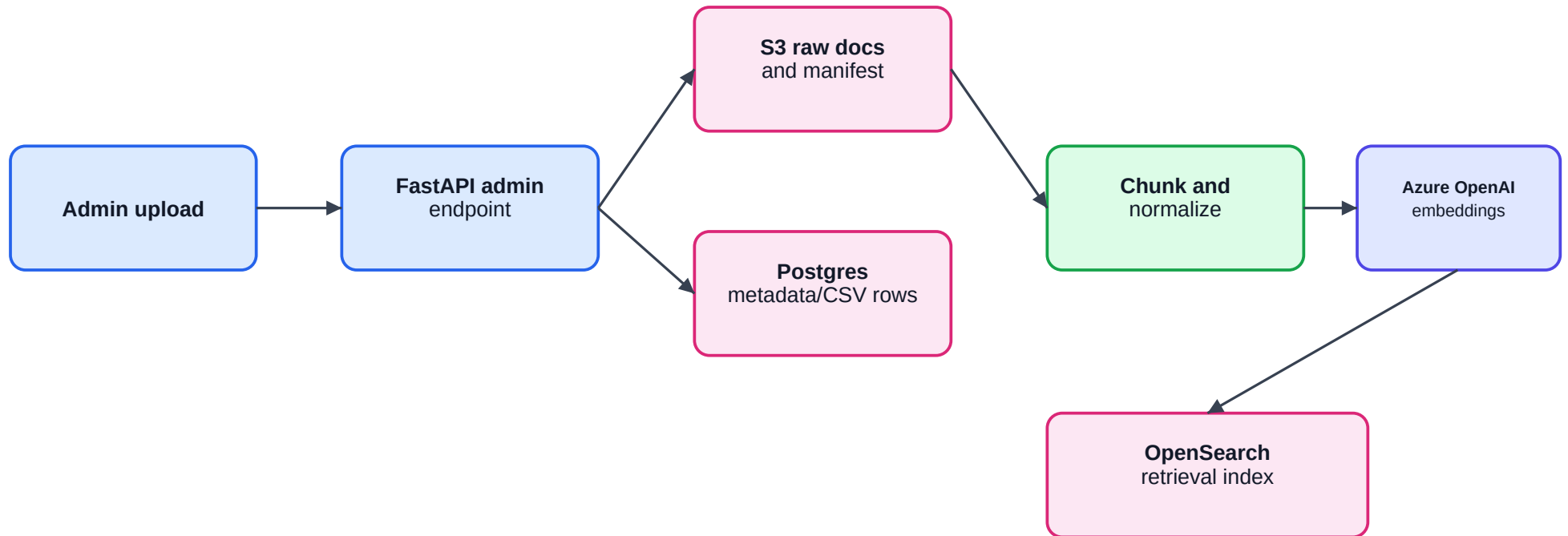
Every online chat request enters the graph. Deterministic lookup is a specialist agent path, not a deterministic preflight shortcut. The supervisor can call multiple specialists before synthesis.

## LLD 2: Deterministic Lookup And CSV Rows



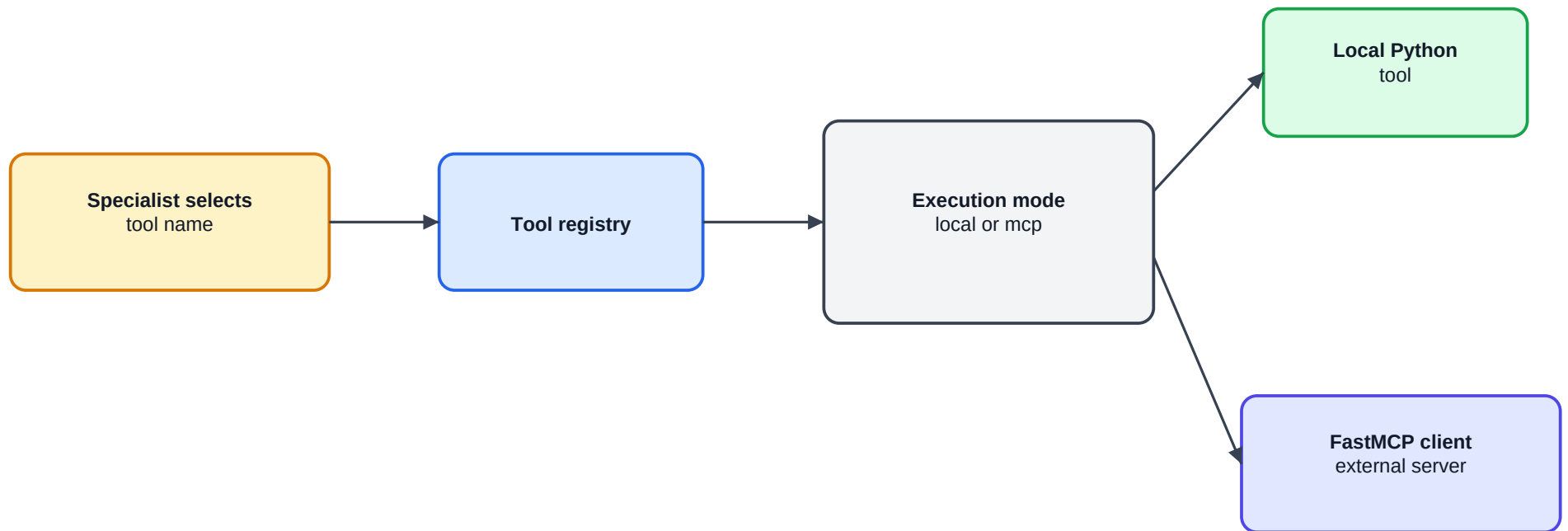
CSV rows are inserted during upload and ingestion. Chat queries read existing PostgreSQL rows only; they do not append rows. On-call questions filter staff schedule rows where `on_call` is Yes. Listing queries return unique instances first, with full detail in an expandable section when needed.

## LLD 3: Document Ingest And Retrieval



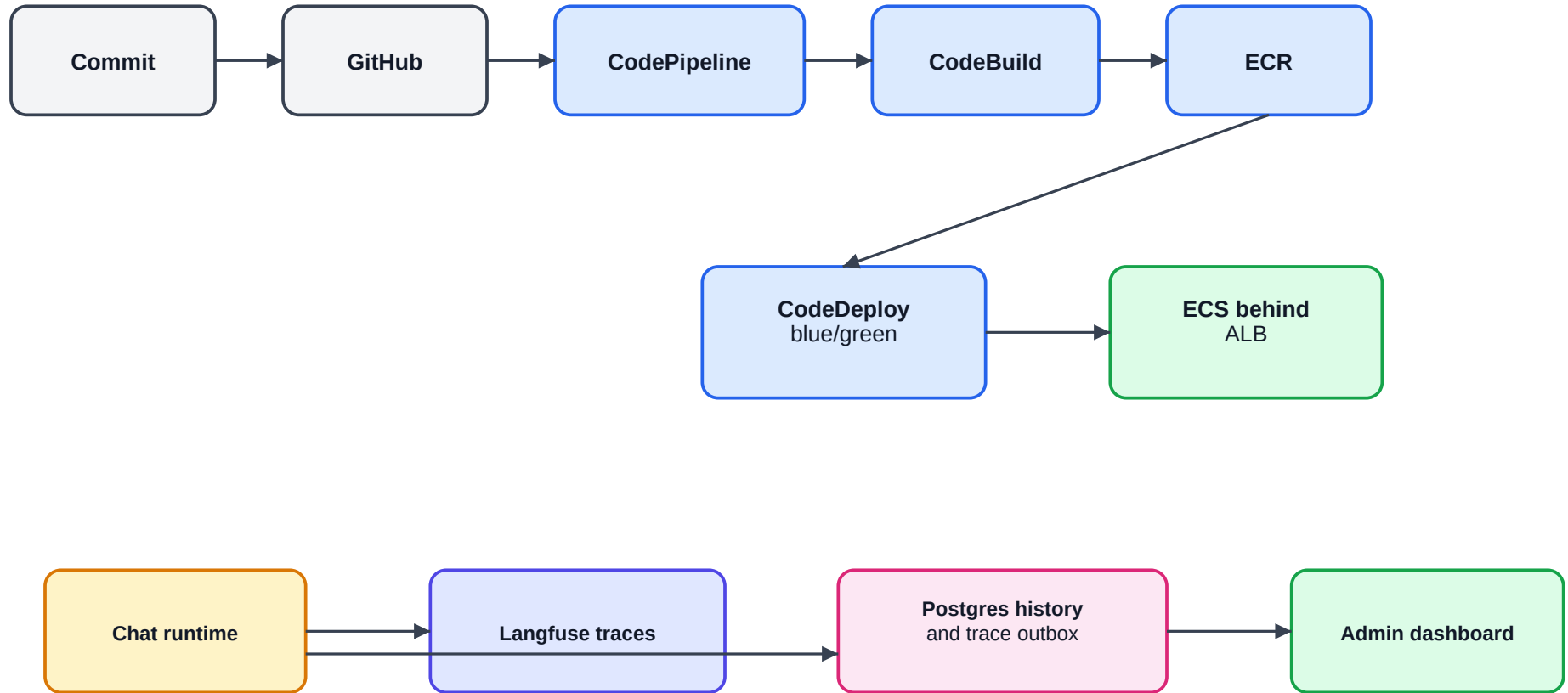
Uploaded documents are stored in S3 and indexed into OpenSearch Serverless. CSV uploads populate deterministic lookup tables in RDS PostgreSQL.

## LLD 4: Tool Execution Boundary



The graph uses stable tool names. `TOOL_EXECUTION_BACKEND` decides whether the implementation runs locally or is forwarded to an external FastMCP tool service.

## LLD 5: CI/CD And Observability



CodeBuild emits rendered ECS task definitions and AppSpec files. CodeDeploy swaps blue/green ECS traffic. Runtime metadata, latencies, tools used, and trace outbox data are persisted for dashboard review and evaluation.